Aikya Sira Kumar

Data & AI Engineer | Data Scientist | ML & MLOps

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Available immediately



Profile

Data scientist with nearly 3 years' experience in analytics and ML, including 9 months at Lufthansa Group (XGBoost, Azure ML MLOps on ~300M booking records). Skilled in GNNs, LLMs/RAG and communicating insights to business teams. English C1, German A2.

Professional Experience

Data Science Intern, *Lufthansa Group* 01/2026 – 09/2026 | Frankfurt, Germany
Project: Advanced Customer Segmentation (ACS)

- Cut the runtime of a quarterly ADF-driven segmentation process by ~30% by building a scalable Azure ML MLOps framework with end-to-end monitoring and governance.
- Reduced Corporate vs. Leisure misclassifications by ~40% compared with a Random Forest baseline by building, tuning and deploying XGBoost and AutoML models on ~300M booking records with engineered ancillary, airport and geospatial features, supporting sales decisions on services and pricing.
- Enabled scalable processing of ~300M booking records by designing Databricks–Data Warehouse pipelines (Delta Tables/Unity Catalog, Azure Data Factory).
- Led a knowledge-transfer session for 4 colleagues on the MLOps pipeline and documented the full process on the internal portal; collaborated weekly with the squad lead (Data & GenAI), sales and network teams.

Senior Data Analyst, *Capgemini Technology Services India Limited* 09/2022 – 09/2024 | Hyderabad, India
Project: Chick-Fil-A

- Supported 99.9% service availability for a global client by building automated KPI monitoring and data validation workflows used by 2–3 teams (~10–15 people).
- Processed operational and web data across 3,000+ locations by developing and optimizing Python/SQL data pipelines.
- Identified operational performance bottlenecks through statistical/KPI analysis and root-cause investigations.
- Translated business requirements into data models and analytical solutions in collaboration with global cross-functional stakeholders.

Projects

Master Thesis – Lufthansa Group, *Airline Network & Passenger Demand Analytics* 04/2026 – 09/2026

- Predicted passenger demand with $R^2 = 0.846$ on a temporally held-out season, generalizing to unseen airports, by building a heterogeneous graph (OAG schedule + IATA DDS ticketing/demand data) and an inductive GraphSAGE model (PyTorch Geometric, custom edge-aware layer).
- Showed that calendar-based features outperform explicit temporal graph structures by benchmarking temporal modeling strategies, and identified structurally critical airports through network vulnerability analysis (articulation points, node-removal simulations).
- Communicated results to technical and non-technical audiences, including the network team that plans schedules and routes, and documented the methodology on the internal portal.

AI Agents 4 Pharma — LLM & AI Agent Development, *Open Source Contributor* 04/2025 – 09/2025

- Resolved 2 high-priority GitHub issues (both merged) and improved prompt logic for autonomous agents in an open-source LLM-based pharma assistant.
- Broadened coverage for biomedical research queries by extending the agent's paper-download capability with bioRxiv and medRxiv alongside arXiv.
- Enabled the agent's retrieval-augmented workflow to use tables and charts from scientific PDFs by designing a multimodal pipeline with NVIDIA NeMo Retriever and PaddleOCR.

NYC Taxi Demand & Weather Impact Analysis, *ELT Pipeline & Tableau BI* 11/2025 – 12/2025

- Identified peak taxi demand on Fridays (4–8 PM) under adverse weather and a strong Manhattan demand concentration by analyzing 1M+ trips in interactive Tableau dashboards (weather-driven demand, hotspots, routes, peak windows).
- Delivered analytics-ready mart tables by building a GCP-based pipeline (monthly trip ingestion, daily weather updates) and transforming raw data with dbt into staging, dimension and fact tables.

EventCast — Real-Time Data Pipeline & ML Application 05/2025 – 06/2025

- Reduced manual event-search effort by ~80% by building a real-time Kafka pipeline (Python microservices, Kafka Connect, MongoDB) ingesting weather and event data from the Open-Meteo, Overpass and Ticketmaster APIs.
- Recommended events across 300+ German cities based on weather preferences with a Streamlit app, powered by a Random Forest model that predicts sunny/shaded conditions on Pandas-cleaned data.

Additional Projects

- NASA Space Apps Challenge 2025 (exoplanet classifier)
- Pacific Dataviz Challenge 2025 (Tableau)
- HR Analytics & Automation System 2025 (Streamlit)

Education

Master of Science in Applied Data Science and Analytics 10/2024 – 09/2026 | Heidelberg, Germany
SRH University
Grade 1.7 (German scale), Thesis: Lufthansa Group

Bachelor of Engineering in Telecommunications Engineering 08/2018 – 07/2022 | Tumkur, India
Sri Siddhartha Institute of Technology
CGPA 8.6/10

Technical Skills

Programming & Data
Python, SQL, Pandas, NumPy, Apache Spark

Deep Learning & AI
PyTorch, TensorFlow/Keras, Graph Neural Networks (GNNs)

Data Engineering
ETL/ELT, Data Pipelines, Data Profiling, Data Validation, Databricks, Delta Lake, dbt

Cloud & Data Platforms
Azure Data Factory, AWS, GCP BigQuery, GCP Vertex AI

Development & Tools
Git, Docker, REST APIs

Machine Learning
Scikit-learn, XGBoost, AutoML, Feature Engineering, Predictive Modeling, Model Evaluation, Statistical analysis

MLOps & Model Deployment
Azure Machine Learning, MLOps, MLflow, Model Monitoring, Data Drift, Monitoring

Generative AI & LLMs
LLM Applications, Prompt Engineering, Agentic AI, Multi-Agent Orchestration, LangChain, NVIDIA NeMo Retriever, FAISS, Embeddings, RAG

BI & Visualization
Tableau, Plotly, Streamlit

Certificates

- Microsoft Certified: Azure Fundamentals (AZ-900)
- Databricks: Deploy Workloads With Lakeflow Jobs
- Fundamentals of Deep Learning (FDL) — NVIDIA Deep Learning Institute (DLI)

Languages

English
C1

German
B1 (Improving Actively)